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Studierichting: Informatica
Oefeningenreeks 5i nr. 1.4

$$f(x) = (2x-1)^{\frac{1}{2}} \text{ op } \left[\frac{1}{2}, 4\right]$$

$$f'(x) = 3(2x-1)^{\frac{1}{2}}$$

$$L = \int_{\frac{1}{2}}^{4} \sqrt{1 + (3(2x-1)^{\frac{1}{2}})^2 dx}$$

$$= \int_{\frac{1}{2}}^{4} (18x-8)^{\frac{1}{2}} dx$$

$$t = 18x - 8 \Rightarrow dx = \frac{dt}{18}$$

$$\Rightarrow \frac{1}{18} \int_{\frac{1}{2}}^{4} t^{\frac{1}{2}} dt = \frac{1}{18} \left[\frac{2t^{\frac{3}{2}}}{3}\right]_1^{64} = \frac{1}{27} \left[t^{\frac{3}{2}}\right]_1^{64}$$

$$= \frac{511}{27}$$

References

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